The dog game with a persistent receiver: results and updates

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Main points. The focus is BIG's averaging receiver in the dynamic Markov game; median-type receivers appear as comparisons.

1. New result. A sticky receiver lands exactly on its rest point in finitely many rounds when an influencer's target decides that point. When a phantom decides it, the receiver only gets closer and never arrives (Proposition 19).

2. In the main game. Averaging arrives exactly when the game has a pivot, with a closed-form number of rounds. About half of random games have one (Proposition 18).

3. The median with inertia. Reporting one's target is not an equilibrium once there is inertia. Reporting what would put the dog on the target next round is one, at every inertia, and that report lies on the line from the dog through the target (Proposition 20).

4. Cartesian and polar moves. Their paths differed because of where the search for a move starts, not because of the movement region: the region accounts for about 3% of the difference (Section 6).

5. New plots. Trajectories and vector fields of the dynamic game (Figs. 1 and 2), and trajectory lines in the live simulator.

1. The game

The Battling Influencers Game (BIG; Wu, Zhu, Cai and Zhu, 2025) has N influencers with targets in the unit square. Each round every influencer reports a point, a receiver (the dog) moves, and each influencer loses the squared distance between the dog and its target, discounted by g per round. Under averaging the dog goes to the weighted mean of the reports, so influencers exaggerate toward the edges. The change here is inertia λ (written w_0 in the code and figures): $x_{t+1} = \lambda\, x_t + (1 - \lambda) \times$ aggregate. The dog's position is then the state of a Markov game. Losses add across coordinates, so the square splits into two one-dimensional games.

Term	Meaning
Target t_i	Where influencer i wants the dog
Report	The point an influencer sends; it need not be its target
Inertia λ	The share of its old position the dog keeps each round
One-shot outcome d*	Where the dog sits in BIG's equilibrium without inertia
Pivot	The one influencer, if any, whose one-shot report is strictly inside the box; it puts the dog on its own target
Phantom	A fixed point added to the reports before taking the median; no influencer controls it
Generalized median	The median of the reports and N - 1 phantoms. Averaging's outcome matches phantoms at k/N; the median has them at the edges
Feedback (Markov perfect) equilibrium	Reports depend only on the dog's position, and at no position can one influencer gain by changing its report
Target form	Each influencer aims the next dog position at a fixed point, as far as it can reach
Rest point	The position where the dog stops moving

2. Trajectories and vector fields of the dynamic game

Figure 1 shows where the dog moves next from every position, and one equilibrium path with every report. Figure 2 shows the influencers' responses. All panels use three influencers at inertia 0.8. Median reports are those of Proposition 20; other equilibrium reports give the same dog path. The figures illustrate one game; the proved statements are in Sections 3 and 5.

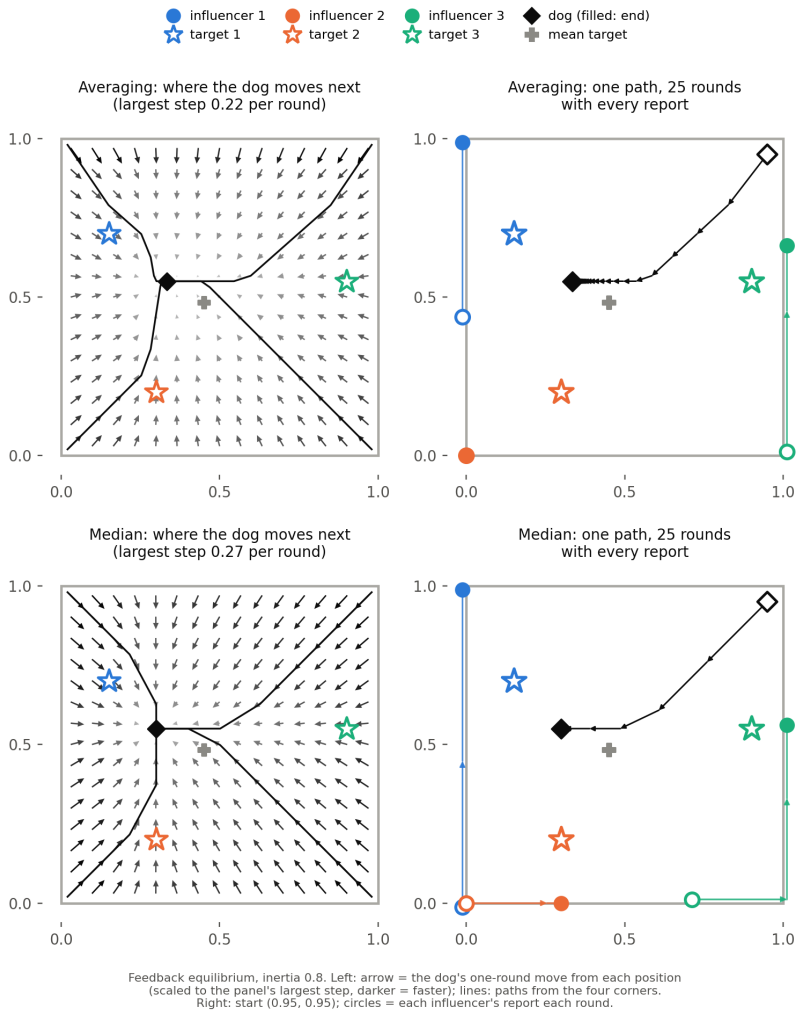


Fig. 1. Left: the dog's one-round move from each position, scaled per panel, with paths from the four corners. Right: one path from (0.95, 0.95) with each influencer's reports. Averaging lands exactly on y = 0.55 in round 5 and only approaches x = 1/3. The median lands exactly in both coordinates, in rounds 6 and 3, and moves faster (largest step 0.27 against 0.22).

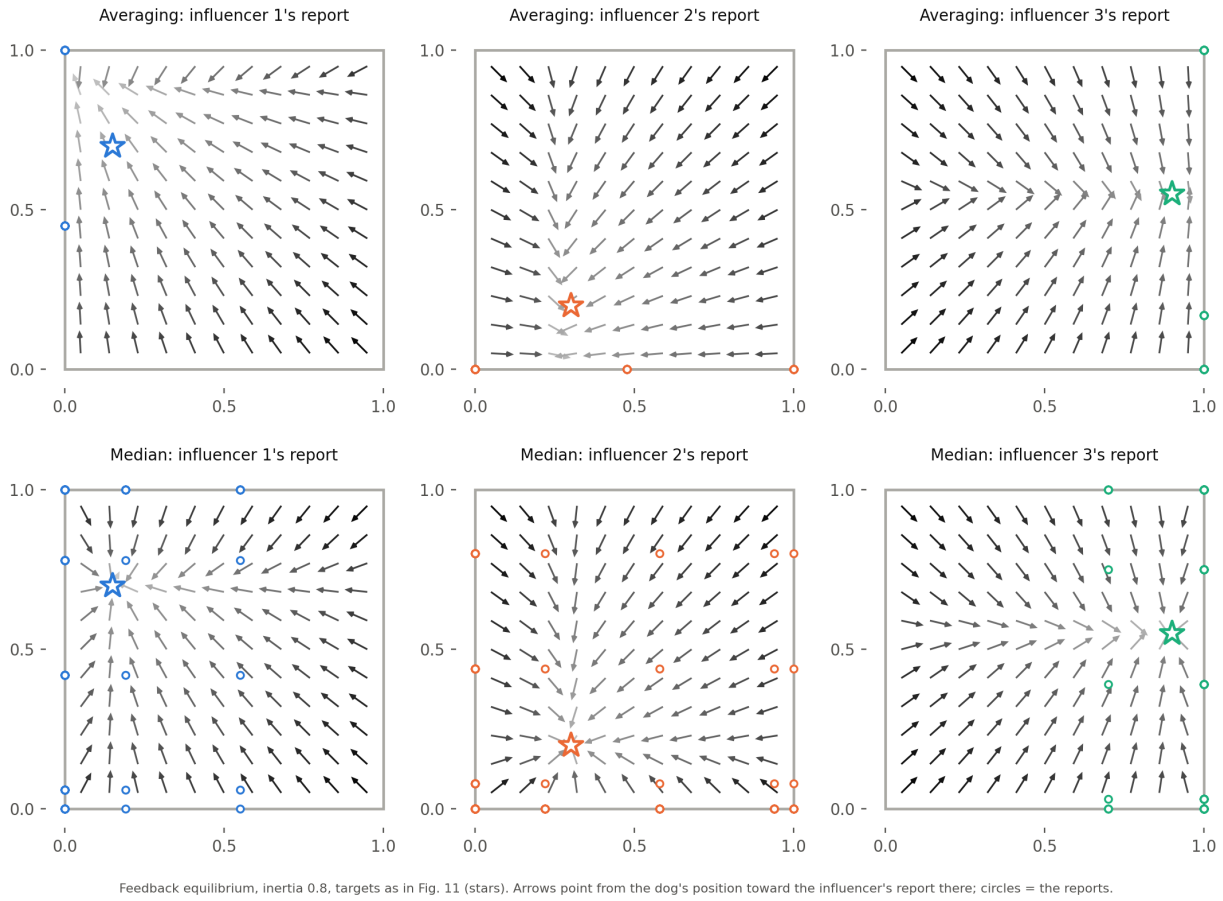


Fig. 2. Response vector fields: at each dog position, an arrow toward the report each influencer makes there in equilibrium (fixed length, darker = farther away), with the reports as circles. Median arrows converge on each influencer's target, because each report lies on the line through it, until a report reaches the edge. Averaging arrows aim at the edges instead, with 2 or 3 distinct reports per influencer.

3. New: which receivers arrive

Proposition 19. Let influencers aim at any targets t_i, with phantoms α_k and $0 < \lambda < 1$, and let m be the generalized median of the targets and phantoms. The receiver's step F is a contraction with modulus λ and its unique rest point is m. If $0 < m < 1$ and m is a target but not a phantom, then $F = m$ on an interval around m, so the receiver reaches m exactly in finitely many rounds. If m is a phantom but not a target, or m is 0 or 1, then $F(x) - m = \lambda(x - m)$ near m, and from any other x the receiver never reaches m.

Proposition 18. With a pivot p, let L and U be the dog with the pivot at 0 and at 1. If edge influencers hold their one-shot reports and the pivot moves the dog as close to its target as it can, the dog reaches d* after exactly $\text{ceil}(\log((U - x)/(U - d^*)) / \log(1/\lambda))$ rounds from $x < d^*$, with the mirror formula above d*. This policy is a feedback equilibrium, at every inertia and discount, if every top influencer has $(t - L)/(1 - L) \geq \lambda$ and every bottom influencer has $(U - t)/U \geq \lambda$.

The reason is simple. A phantom is fixed, so inertia shrinks the gap by a factor λ every round. An influencer's report is recomputed every round to make up for inertia, so a pivotal influencer cancels it and lands the dog on its target.

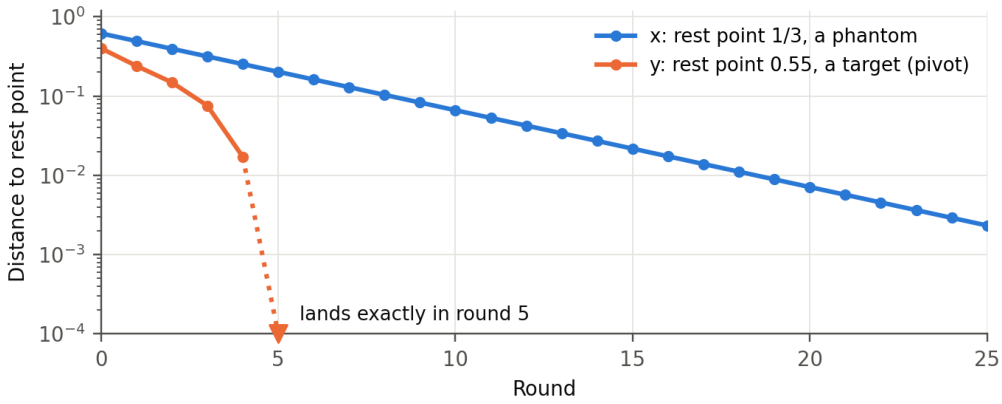


Fig. 3. Distance to the rest point for the averaging receiver of Fig. 1 (log scale). The x-coordinate, decided by a phantom, falls by a factor 0.8 per round and is still 0.0023 away at round 25. The y-coordinate, decided by a target, reaches 0.

Receiver	Exact arrival?	Why
Averaging	Only with a pivot	Otherwise the rest point is a phantom k/N
Median (odd N)	Yes, unless the median target is 0 or 1	The rest point is the median target
Compensated	Yes, for any interior rest point	It clips the whole aggregate to what it can reach

Evidence. Both propositions have written proofs and tests. Proposition 19 also held in 3,000 random cases. In random games a pivot appeared in 115, 96 and 100 of 200 games for 3, 5 and 7 influencers. In all of those games the target-form path arrived, with a median of 3 to 23 rounds from the worst start (inertia 0.5 to 0.9). **Limits.** The condition in Proposition 18 is sufficient, not necessary, and is rarely met at high inertia: it held in 52 of 115 pivot games at three influencers and inertia 0.5, and in at most 11 of about 100 in every other setting. When it fails, edge influencers usually gain by easing off (up to 0.077). Proposition 19 is about paths and does not say that a given set of targets is an equilibrium.

4. The feedback equilibrium of the averaging game

- **Proposition 1.** Holding the one-shot actions at every position is an equilibrium if and only if every edge influencer's slack $(t - d^*)/(1 - d^*)$ is at least $\rho(\lambda, g) = \lambda(1 - g\lambda)/(1 - g\lambda^2)$, and no one-shot action is inside the box.
- **Proposition 2.** With two players whose targets straddle d^*, each player aiming the next position at $d^* + k(t - d^*)$, with $k = (1 - g\lambda^2)/(1 - g\lambda)$, is an equilibrium at every inertia and discount. It was an equilibrium in 77 of 77 random straddling games.
- **Overshooting.** Players aim past their own targets. The overshoot factor k is largest at an intermediate inertia, where it is at most 2.
- **The rest point does not depend on inertia.** In Propositions 1, 2 and 18 the dog ends at the one-shot outcome d^* at every inertia, and the long-run losses in the saved random games do not change with inertia. Inertia changes how fast the dog gets there.

5. The median with inertia, and how its directions differ from averaging

Proposition 20. For an odd number of influencers with distinct targets, at every inertia and discount, the reports $r_i(x) = \text{clip}((t_i - \lambda x)/(1 - \lambda), 0, 1)$ are a feedback equilibrium, with the same dog path as Proposition 4. Unclipped, $r_i = t_i + (\lambda/(1 - \lambda))(t_i - x)$: the report lies on the line from the dog through the target, $1/(1 - \lambda)$ times as far. A report never points away from the target. With discount 0 the report is weakly dominant.

Is truthful reporting the median's equilibrium? Reporting the target itself is an equilibrium only without inertia. In the game of Fig. 1, an influencer gains 0.03 to 0.11 by deviating from it at inertia 0.3 to 0.9, and 0 at inertia 0 (Proposition 3). The dog moves only part of the way to the median, so a report of t leaves it short of t. What survives is truthfulness about the next position: each influencer reports what would put the dog on its target next round. That profile had no profitable deviation in any of 42 random games with 3, 5 and 7 influencers. Reports are not unique: only the pivotal influencer's report is pinned down, and the others need only stay on the same side of it, so an earlier version of Fig. 2 showed equally valid reports at the edges, with the same dog path.

Why the directions differ. Under averaging every report adds to the outcome, so an influencer pushes as hard as it can and only the side matters; reports sit at the edges, and an influencer can keep pulling away from its own target to offset the others. Under the median, pushing past the pivot adds nothing, so a report only has to be right, and it aims at the target. The gap closes with inertia, because the factor $1/(1 - \lambda)$ sends median reports to the edges too.

Inertia	Angle from report to target: averaging / median	Coordinates pointing away from the target: averaging / median	Reports inside the box: averaging / median
0.2	13.1° / 0.0°	16.7% / 0%	17.5% / 87.3%
0.5	15.4° / 4.0°	14.3% / 0%	9.5% / 50.0%
0.8	16.5° / 11.6°	11.9% / 0%	4.0% / 23.0%
0.9	16.5° / 15.0°	10.3% / 0%	2.4% / 11.1%

The game of Fig. 1, over a grid of dog positions (median angle). In 100 random three-influencer games at inertia 0.8 the means are 14.2° / 10.0°, 10.4% / 0% and 4.7% / 20.1%.

Which receiver loses less. Without inertia averaging has the lower social loss in most random games, and the median wins in about a fifth. At inertia 0.9 the median wins in 81 to 88% of games, for 3, 5 and 7 influencers (numerical, 100 games per point). The cause is speed, not location: long-run outcomes do not change with inertia, but under averaging influencers on the far side keep pulling, while the median moves at full speed once a majority pushes one way.

6. Why Cartesian and polar moves gave different paths

Continuous-move Nash-Q reached the Nash corners with (vx, vy) moves and not with (r, θ) moves, in the same game. The earlier runs could not say why: both used the same disk of moves. A new 2 × 2 × 2 study varied the network's input, the coordinates of the search for a move, and the region, disk or square (10 seeds per cell, 80 runs).

Source of the difference between mean paths	Share
Search coordinates	74.0%
Search × network input	12.3%
Network input	10.9%
Region (disk or square), with its interactions	2.8%

- **Region.** Cartesian and polar paths are 0.403 apart on the disk and 0.406 on the square. Disk and square are 0.011 apart with Cartesian moves and 0.030 with polar moves. So the region is about 3 to 7% of the gap. Without any learning they differ by 0.034: a square lets a player reach a corner in 5 rounds instead of 7.
- **Search.** Replaying the same trained networks with a grid search cuts the gap from 0.403 to 0.048. The polar networks had learned nearly the right values, and the search could not find the moves they imply.
- **Cause.** The search starts both players at $\theta = 0$. Blue's corner is to the north-west, and blue stays 0.46 from it. Starting both at $\theta = \pi$ flips the failure to red (0.41). Starting each on its own side works without retraining.
- **Fix.** Trying 8 directions before the gradient steps, in training and in play, brings all 10 seeds within 0.1 of the corners and the paths to 0.034 from Cartesian. This is one game, one step length and one network size.

7. Other results, in brief

- **Compensated receivers (side thread).** A receiver that moves to the reachable point closest to a strategy-proof rule's outcome keeps literal truthful reporting an equilibrium (Theorem 1). It needs the receiver to be able to stay put and its reach not to fall (Propositions 15 and 16). Its welfare has a closed form (Theorem 2). Its price of anarchy never rises with inertia (Proposition 14). Symmetric noise keeps truthful reporting (Proposition 17).
- **Learning.** Self-play finds the one-shot equilibrium: 74 of 75 REINFORCE runs and 25 of 25 PPO runs. With a persistent dog, A2C and PPO hold the one-shot actions in 19 of 20 runs and miss the feedback equilibrium. Fitting each state's exact best response brings the largest gain from 0.048 and 0.104 to at most 0.001 (5 of 5 seeds).

8. Updates and corrections

- **New:** Propositions 19 and 20, the study of Section 6, redrawn median reports in Figs. 1 and 2, and trajectory lines with kept runs in the live simulator.
- **Polar moves.** The earlier statement that polar moves never reach the corners held only for a search that starts from one fixed angle.
- **Theorem 3 narrowed.** The best compensated phantoms do not depend on inertia when each phantom's criterion changes sign once, as for uniform targets. With clustered targets inertia can switch between static local optima.
- **Number corrected.** The gain from shading under clipped shocks is about 0.0015 at inertia 0.8, not 0.0024, and about 0.0001 without inertia, not 0.0003. The old values were quadrature error.
- **Literature.** Wada et al. (AAMAS 2018) prove a special case of Theorem 1, and Jennings et al. (2021) give the static version of Theorem 3.

9. Open problems and how new this is

- **General pivot case.** Without the condition of Proposition 18 the equilibrium is not proved. Numerically, target form is an equilibrium in 10 of 12 checked games. In the other 2 (five influencers, inertia 0.9), an influencer gains up to 0.005, because its one-round objective has several peaks. Value iteration cut the gain to 0.001, limited by its grid.
- **Not covered:** uniqueness of the feedback equilibrium, unequal weights in Proposition 19, and a rest point that is both a target and a phantom. Section 6 did not separate a local peak from too few search steps.
- **Novelty.** As far as the searches went, no earlier work combines strategic reports, a persistent median-type receiver and this split between targets and phantoms. Weighted-median opinion dynamics (Mei et al., 2022) has finite-time convergence, and Cheng and Mei (2026) study an inertial median, but their agents are not strategic. The searches were not a systematic review, and the mathematics of Propositions 19 and 20 is elementary, so it may appear elsewhere in another form. None of the proofs has been refereed.